

DIFFERENTIALS, DUAL ISOGENIES, AND THE TATE MODULE

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Three computations by hand, on the invariant differential, the dual isogeny and the Tate module, each comparing the multiplicative group $\mathbb{G}_m$ with the elliptic curve

$$E: y^2 = x^3 - x, \quad \omega_E := \frac{dx}{2y}.$$

Throughout, k is a field with $\text{char } k \neq 2$, $\bar{k}$ an algebraic closure and $k^{\text{sep}} \subseteq \bar{k}$ the separable closure; $\mathbb{G}_m = \mathbb{P}_k^1 \setminus \{0, \infty\}$ has coordinate t and invariant differential $\omega_{\mathbb{G}_m} := dt/t$.

Reading. [Sil09, §§III.4–III.7], and [Sil09, §III.8] for the Weil pairing of Fact 3.1.

A formula. We use the following duplication formula on E , derived in Problem 1.1(5): if $x \in k(E)$ is the x -coordinate and $[2]: E \rightarrow E$ is multiplication by 2, then

$$(0.1) \quad [2]^*x := x \circ [2] = \frac{(x^2 + 1)^2}{4(x^3 - x)}.$$

1. INVARIANT DIFFERENTIALS

Problem 1.1 (Poles and pullbacks).

- (1) Rewrite $\omega_{\mathbb{G}_m}$ in the coordinate $s := 1/t$ near ∞ , and compute $\text{div}(\omega_{\mathbb{G}_m})$ on $\mathbb{P}_k^1$. Compute $[m]^*\omega_{\mathbb{G}_m}$ for $[m](t) = t^m$, $m \in \mathbb{Z}$.
- (2) Show that $\omega_E = dy/(3x^2 - 1)$, and that y and $3x^2 - 1$ have no common zero on E . Deduce that ω_E has no zeros or poles on $E \setminus \{O\}$.
- (3) Using $\text{ord}_O x = -2$ and $\text{ord}_O y = -3$, show that $\text{ord}_O \omega_E = 0$.
- (4) Compute $[-1]^*\omega_E$ from $[-1](x, y) = (x, -y)$, and compare with $[-1]^*\omega_{\mathbb{G}_m}$.
- (5) We derive (0.1) by hand. Here $x, y \in k(E)$ are functions; for a point $P \in E(\bar{k})$ write $x_P := x(P)$ and $y_P := y(P)$ for its coordinates.
 - (a) Let $P \in E(\bar{k})$ with $y_P \neq 0$. Show that the tangent line to E at P has slope $\lambda_P = (3x_P^2 - 1)/(2y_P)$, and use that its third point of intersection with E is $-[2]P$ to show that $x([2]P) = \lambda_P^2 - 2x_P$.
 - (b) Let $\lambda := (3x^2 - 1)/(2y) \in k(E)$, so that $\lambda(P) = \lambda_P$. Explain why (5)(a) gives the identity $[2]^*x = \lambda^2 - 2x$ of functions in $k(E)$, and simplify it using $y^2 = x^3 - x$.
- (6) Solve this part in **Magma**, working in the function field $\mathbb{Q}(E) = \mathbb{Q}(x)[y]/(y^2 - x^3 + x)$. Compute $[2]^*x$ and $[2]^*y$ from the function λ of (5), check (0.1), and show that

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3 - x)}, \quad g_0 = x^2 + 1, \quad g_1 = x^2 - 2x - 1, \quad g_{-1} = x^2 + 2x - 1,$$

for $e \in \{0, 1, -1\}$. Show that $[2]^*y = \pm(x^2 + 1)(x^4 - 6x^2 + 1)/(8y^3)$ and $[2]^*\omega_E = \pm 2\omega_E$. Which sign is the correct one?

2. DUAL ISOGENIES

Consider the elliptic curves

$$E_1: y_1^2 = x_1^3 - x_1, \quad E_2: y_2^2 = x_2^3 + 4x_2, \quad E_3: y_3^2 = x_3^3 - 16x_3,$$

so that $E_1 = E$, and write $\omega_i := dx_i/(2y_i)$ for the invariant differential of E_i . Consider the maps

$$\begin{aligned} \varphi: E_1 &\rightarrow E_2, & (x_1, y_1) &\mapsto \left(\frac{x_1^2 - 1}{x_1}, \frac{y_1(x_1^2 + 1)}{x_1^2} \right), \\ \psi: E_2 &\rightarrow E_3, & (x_2, y_2) &\mapsto \left(\frac{y_2^2}{x_2^2}, \frac{y_2(x_2^2 - 4)}{x_2^2} \right), \\ \iota: E_3 &\rightarrow E_1, & (x_3, y_3) &\mapsto \left(\frac{x_3}{4}, \frac{y_3}{8} \right). \end{aligned}$$

For an isogeny $\alpha: E_i \rightarrow E_j$, define c_α by $\alpha^*\omega_j = c_\alpha\omega_i$. The maps φ and ψ are special cases of [Sil09, Example III.4.5], which treats $y^2 = x^3 + ax^2 + bx$ in general, so they are isogenies; ι is an isomorphism. That example computes the kernels and the dual, but not the constants c_α : that is part (2) of Problem 2.1.

Problem 2.1 (A 2-isogeny and its dual).

- (1) Check that φ lands on E_2 . Show that $\ker \varphi = \{O_1, (0, 0)\}$, that $\varphi(\pm 1, 0) = (0, 0)$, and that $\deg \varphi = 2$.^a
- (2) Compute c_φ .
- (3) Check that ψ lands on E_3 and that ι lands on E_1 , and compute c_ψ and c_ι .
- (4) Show that $(\iota \circ \psi \circ \varphi)^*x_1 = [2]^*x_1$, so that $\iota \circ \psi \circ \varphi = \pm[2]$. Use (2) and (3) to decide the sign, and conclude that $\hat{\varphi} = \iota \circ \psi$, that $c_\varphi c_{\hat{\varphi}} = \deg \varphi$, and that $\ker \hat{\varphi} = \varphi(E_1[2](\bar{k}))$.
- (5) Is there an isogeny $\mathbb{G}_m \rightarrow \mathbb{G}_m$ of degree 2? If so, does it have a dual?

^aHint: write y_1 in terms of φ^*y_2 and x_1 , and find a quadratic equation for x_1 over $\varphi^*k(E_2)$.

3. THE TATE MODULE

We first describe the Tate module of $\mathbb{G}_m$, the familiar case, and then repeat each step for E . What links the two is the Weil pairing, which you may use without proof.

Fact 3.1 (The Weil pairing). Let $m \geq 1$ with $\text{char } k \nmid m$. There is a pairing

$$e_m: E[m](k^{\text{sep}}) \times E[m](k^{\text{sep}}) \rightarrow \mu_m(k^{\text{sep}})$$

that is bilinear, alternating ($e_m(P, P) = 1$), non-degenerate, and Galois equivariant: $\sigma(e_m(P, Q)) = e_m(\sigma P, \sigma Q)$ for every $\sigma \in \text{Gal}_k$ [Sil09, Proposition III.8.1]. Here $E[m](k^{\text{sep}}) = E[m](\bar{k})$, since $\text{char } k \nmid m$. It follows that if (P, Q) is a basis of $E[m](k^{\text{sep}})$ over $\mathbb{Z}/m\mathbb{Z}$, then $e_m(P, Q)$ is a primitive m -th root of unity, and if σ acts on this basis by the matrix $A \in \text{GL}_2(\mathbb{Z}/m\mathbb{Z})$, then $\sigma(e_m(P, Q)) = e_m(P, Q)^{\det A}$.

Problem 3.2 (The 2-power torsion of $y^2 = x^3 - x$).

Regard E as a curve over $\mathbb{Q}$, fix $\bar{\mathbb{Q}} \subset \mathbb{C}$, and recall that $E[m](\bar{\mathbb{Q}})$ is the kernel of $[m]$ on $E(\bar{\mathbb{Q}})$. Let σ be complex conjugation, acting on $\bar{\mathbb{Q}}$ and on $E(\bar{\mathbb{Q}})$ coordinatewise.

- (1) Recall that $T_2(\mathbb{G}_m) := \varprojlim_n \mu_{2^n}(\bar{\mathbb{Q}})$, with transition maps $\zeta \mapsto \zeta^2$. Find a $\mathbb{Z}_2$ -basis of $T_2(\mathbb{G}_m)$, and write down its first three terms. How does σ act on $\mu_4(\bar{\mathbb{Q}})$, and on $T_2(\mathbb{G}_m)$?

- (2) List $E[2](\overline{\mathbb{Q}})$. Using Problem 1.1(6), find the x -coordinates of all $P \in E(\overline{\mathbb{Q}})$ with $[2]P \in E[2](\overline{\mathbb{Q}}) \setminus \{O\}$, and conclude that $\#E[4](\overline{\mathbb{Q}}) = 16$.
- (3) Show that $P_2 := (i, 1-i)$ and $Q_2 := (1+\sqrt{2}, 2+\sqrt{2})$ are points of $E(\overline{\mathbb{Q}})$, with $[2]P_2 = (0, 0)$ and $[2]Q_2 = (1, 0)$. Show that (P_2, Q_2) is a basis of $E[4](\overline{\mathbb{Q}})$ over $\mathbb{Z}/4\mathbb{Z}$, and that $E[4](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q}(\zeta_8))$. Compare with $\mu_8(\overline{\mathbb{Q}})$ and Fact 3.1.
- (4) Show that σ restricts to a group automorphism of $E[4](\overline{\mathbb{Q}})$. Compute $P_2 + (1, 0)$, and write $\sigma(P_2)$ and $\sigma(Q_2)$ in the basis (P_2, Q_2) . Compare the determinant of the resulting matrix in $\text{GL}_2(\mathbb{Z}/4\mathbb{Z})$ with the action of σ on $\mu_4(\overline{\mathbb{Q}})$ found in (1), and explain the agreement using Fact 3.1.
- (5) Let $(P_n)_n$ and $(Q_n)_n$ be compatible sequences, $[2]P_{n+1} = P_n$ and $[2]Q_{n+1} = Q_n$, starting with $P_1 = (0, 0)$, $Q_1 = (1, 0)$ and P_2, Q_2 as above. Explain why they form a $\mathbb{Z}_2$ -basis of $T_2(E)$, and compare with (1).

4. MAGMA EXAMPLES

Example 4.1 (The isogenies of Problem 2.1). We check in the function field of E_1 that φ and ψ land where they should, that $\iota \circ \psi \circ \varphi = [2]$, and compare with Magma's own 2-isogeny.

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E := EllipticCurve([-1,0]);           // y^2 = x^3 - x
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>;      // x = x_1, y = y_1
x2 := F!((x^2 - 1)/x); y2 := y*(x^2 + 1)/x^2; // phi^*x_2, phi^*y_2
y2^2 eq x2^3 + 4*x2;                  // true
x3 := y2^2/x2^2; y3 := y2*(x2^2 - 4)/x2^2; // (psi o phi)^*x_3, ^*y_3
y3^2 eq x3^3 - 16*x3;                  // true
lam := (3*x^2 - 1)/(2*y); xd := lam^2 - 2*x; yd := -(y + lam*(xd - x));
x3/4 eq xd and y3/8 eq yd;             // true: [2]
phi := TwoIsogeny(E![0,0]);
aInvariants(Codomain(phi));            // [ 0, 0, 0, 4, 0 ]
phi(E![1,0]);                          // (0 : 0 : 1)
DualIsogeny(phi);
// ... taking (x : y : 1) to ((1/4*x^2 + 1) / x : (1/8*x^2*y - 1/2*y) / x^2 : 1)

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Magma's dual is $\iota \circ \psi$ written out in the coordinates x_2, y_2 of E_2 : $((x_2^2 + 4)/(4x_2), y_2(x_2^2 - 4)/(8x_2^2))$.

Example 4.2 (Division polynomials and $E[4](\overline{\mathbb{Q}})$). For $m \geq 2$, $\text{DivisionPolynomial}(E, m)$ vanishes exactly at the x -coordinates of $E[m](\overline{\mathbb{Q}}) \setminus \{O\}$. Its degree is $(m^2 - 1)/2$ for odd m and $(m^2 + 2)/2$ for even m , and its factors for $m = 4$ are $x, x \pm 1$ and the g_e of Problem 1.1(6).

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[ Degree(DivisionPolynomial(E, m)) : m in [2..8] ];
// [ 3, 4, 9, 12, 19, 24, 33 ]
Factorization(DivisionPolynomial(E, 4));
// x - 1, x, x + 1, x^2 - 2*x - 1, x^2 + 1, x^2 + 2*x - 1
Q8<z> := CyclotomicField(8); i := z^2; s := z + z^7; // s^2 = 2
E8 := BaseChange(E, Q8);
P2 := E8![i, 1-i]; Q2 := E8![1+s, 2+s];
2*P2, 2*Q2; // (0 : 0 : 1) (1 : 0 : 1)
#DivisionPoints(E8!0, 4); // 16
E8![-i, 1+i] eq 3*P2 + 2*Q2; // true: complex conjugation

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Example 4.3 (Characteristic p). Reducing modulo p , the p -division polynomial collapses, because $\#E[p](\overline{\mathbb{F}}_p) \in \{1, p\}$.

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for p in [3,5,7,13] do
  Ep := ChangeRing(E, GF(p));
  psi := DivisionPolynomial(Ep, p);
  print p, IsSupersingular(Ep), Degree(psi), Factorization(psi);
end for;
// 3 true 0 []
// 5 false 10 [ <x^2 + 2, 5> ]
// 7 true 0 []
// 13 false 78 [ <x^6 + 11*x^4 + 5*x^2 + 11, 13> ]

```

For $p = 3, 7$ the curve is supersingular and $E[p](\overline{\mathbb{F}}_p) = 0$; in general, $y^2 = x^3 - x$ is supersingular exactly when $p \equiv 3 \pmod{4}$. For $p = 5, 13$ the division polynomial is the p -th power of a polynomial of degree $(p-1)/2$, whose roots are the x -coordinates of the $p-1$ non-zero points of $E[p](\overline{\mathbb{F}}_p) \cong \mathbb{Z}/p\mathbb{Z}$; the exponent p is the inseparable degree of $[p]$. The same computation for $\mathbb{G}_m$ is $t^p - 1 = (t - 1)^p$.

Looking ahead. The Weil pairing of Fact 3.1 also gives $\det \varphi_\ell = \deg \varphi$ for the map φ_ℓ induced by an isogeny on Tate modules ([Sil09, Proposition III.8.6]); formal groups make c_φ an honest derivative; and Frobenius factors $[p]$ in characteristic p .

REFERENCES

- [Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑